

# Sub-word formation and the historical development of number in Semitic

**Aim** In the theoretical context of Distributed Morphology, we proposed in earlier work that the Arabic gender suffix  $-a(t)$ , originally a derivational suffix, underwent a series of internal and external changes to become an inflectional morpheme, adding evidence for the idea that historical changes are not limited to changes in the status of linguistic terminals from M-word to Subword or vice versa (Roberts and Roussou, 1999, 2003; Van Gelderen, 2011), but that they can also occur *within* an M-word (see Diertani 2011). In this paper, we pursue further the idea that there can be internal changes in M-words, targeting sub-words, by focusing on the evolution of the plural in Semitic, and in Arabic in particular. We focus on the development of the external (sound, suffixal) plural, but also discuss the case of the internal (broken) plural. Our approach is comparative and deductive, and based on theoretical insights from Distributed Morphology as well as featural accounts of number (Noyer, 1992; Nevins, 2011; Harbour, 2011, 2014).

**Gender** In earlier work, we showed that the feminine marker  $-a(t)$  in Arabic is used, not only as a gender marker as in (1), but also to produce nouns from adjectives (2), collectives from singulars (3) and singulatives from collectives (4). That such nouns are all marked feminine is somehow strange. For example, why should ‘butchers’ become “feminine” when viewed as a group? Diachrony provides us with clues for this state of affairs: there is evidence from reconstruction studies (Hasselbach, 2014a,b) that  $-a(t)$  was originally a derivational morpheme and that it was only secondarily associated with feminine gender. On this view, the original function of the suffix was to mark derivatives of adjectives, specifically abstract nouns (2) and other usages of  $-a(t)$  derived from this basic function whose structure is in (5). One of the common sources of morphosyntactic change is a misunderstanding by language learners of which structural position an exponent is associated with (Diertani, 2011). We proposed this is what happened with  $-a(t)$ . A derivational morpheme in  $n$  was reanalysed as an inflectional morpheme in Num, as shown in (6). This is a change of function involving a Subword. We propose further that, via a novel system of polarity (Hasselbach, 2014b), Arabic  $-a(t)$  came to denote groups under Num, as shown in (7). Since there is evidence that feminine gender was first marked on adjectives and was marked only secondarily on nouns (Hasselbach, 2014b), we proposed that  $-a(t)$  was reanalysed in Num as a gender marker via the addition of a feature [u +Fem], as in (8). This feature spread to D where it could enter in an agreeing relationship with other elements (this was probably due anaphoric pressures), but this was made possible, we claim, because Arabic had acquired a polarity system where  $-a(t)$  could act on a given input with various results. Via a system of contrast, a neutral/masculine noun under  $n$  was made feminine via agreement under Num. The last step involved the spreading of [u +Fem] on  $n$ , as in (9). This involves a Subword changing its function from inflectional to derivational.

**Number** In this paper, we focus on the development of number in Semitic. Following Lipiński (2001) and Haelewyck (2016), we propose that originally in Arabic “the singular was the only number, while the plural was expressed by the collective singular, characterized in Semitic by a different pattern of the basic root, corresponding to the so-called “broken plural” [...] The plural distinguished by additional morphemes from the singulars and from the collective singulars is then the result of a further development.” (Lipiński, 2001, p. 236). We note that collectives, e.g. *toffeḥ* ‘apple’, semantically plural nominals can have the same patterns as singulars, e.g. *falleḥ* ‘farmer’, suggesting that they are singular morphosyntactically (they also take third person singular agreement) and broken plurals, e.g. *kilaab* ‘dogs’, can also share patterns with singulars, e.g. *kitaab* ‘book’, suggesting broken plurals are underlyingly singular (they can agree with verbs in the singular). Suppose first that collective nouns are generated under  $n$ , as in (6), broken plurals start appearing when internal change occurs and at first broken plurals are under  $n$ , but gradually broken plurals systematically and productively denote plurality, becoming inflectional, and a change from  $n$  to Num is happening, as in (10). This is a mutation from a derivational to an inflectional morpheme (stem internal). On our view, the broken plural is thus not an innovation in the sense that it appears after the suffixal plural, as has been claimed in the literature, see Ratcliffe (1998) (for ex. broken plurals are non-existent in Hebrew). The next puzzle is how the external (i.e. suffixal)

plural developed. We propose that the first suffixal marking for number was the paucal. This paucal number being an operation on a singulative in Num, another Num projection is required, as in (11). To this day, the suffix *-a(t)* is ambiguous in Arabic, it can denote plurality or paucality. We speculate that paucal *-a(t)* started to be used when Arabic lost its paucal system from 3 to 10: those numbers were marked by a special affix? (Haelewyck, 2016). The next step involves lowering to the first Num projection denoting plurality, as in (12). The suffix *-a(t)* started denoting ‘more than one’. Here we have a change from one inflectional node to another. Presumably, by analogy, a masculine suffixal plural, i.e. *een*, was created. Broken plurals and collectives (as well as singulatives) continued to exist. Arabic has now two systems: one like English and French based on a singular/plural contrast and one based on a collective/singular contrast. The latter is the archaic system whose residues are still pervasive in Semitic grammars.

- (1) ameer                      ameer-a(t)  
prince.MASC.SG   prince-FEM.SG  
‘prince, princess’

- (2) safeed    safad-a(t)  
happy    happiness  
‘happy, happiness’

- (3) jazzar                      jazzar-a(t)  
butcher.MASC.SG   butcher-pl  
‘butcher, butchers’

- (4) beeð            beeð-a(t)  
egg.COLL    egg-SING  
‘eggs, an egg’

